

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Computer Science & Engineering

Simulation of Domain Name Resolution Using UDP Socket Programming

Deadline: 05/08/26

Develop a client/server program using UDP to simulate the DNS server and the DNS client. The client will query the DNS server for domain name resolution and the server respond with its corresponding IP address. Also, analyse the suitability of UDP for DNS services in terms of communication overhead, latency, and reliability. (CO2, K3, 1.4.1, 2.1.3, 2.3.1)

Requirements:

- DNS Database: A dictionary acting as the DNS table (hostname → IP).
- Name Resolution Function: Query a domain name and get its corresponding IP.
- Caching: Simulate caching to store recent lookups in the client side
- Error Handling:
 - Domain not found (NXDOMAIN) -> non-existent domain
 - Timeout simulation
 - Invalid format